

# OpenClaw AI Script Writing for Peripheral Control Tutorial

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Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on "letting AI generate independently runnable scripts".

## 1. Tutorial Objectives

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This article demonstrates through "natural language generation hardware scripts" that OpenClaw can not only directly control peripherals, but also help you save control logic as script files. After completing this tutorial, readers can:

- Understand the basic approach to OpenClaw generating hardware control scripts
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code
- Generate four common script types: RGB, servo, temperature and humidity, OLED

## 2. Hardware Preparation

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- OpenClaw mainboard
- Servo / RGB light strip / OLED / DHT temperature and humidity sensor
- `[Optional] AI voice module + speaker`

Description:

- This tutorial does not require connecting all peripherals at once
- Connect the corresponding hardware first for the type of script you want to demonstrate

## 3. Software Preparation

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**Code entry points that will be used in the tutorial**

- Script generation skill: `/home/jetson/.openclaw/workspace/skills/script-runner/SKILL.md`
- Default script save directory: `/home/jetson/Desktop/`
- Hardware command example directory: `/home/jetson/hardware_ha1_demo/`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The core of this type of capability is not "executing an immediate command", but letting OpenClaw help you generate an independently runnable Python script. According to existing skill agreements:

- When users say "generate script", "write a script", "save to desktop", they will prioritize using `script-runner`
- Hardware scripts use `openclaw_lib` by default
- Default serial port is `/dev/ttyTHS1`

- Default baud rate is 115200
- Scripts are usually saved in /home/jetson/Desktop/

You can understand it as two usage methods:

- Method one: Generate script and run immediately
- Method two: Only generate script, save to desktop first, modify or run again later

## 4. Test Script Generation Capability

Before starting this tutorial, please complete one `openclaw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After confirming OpenClaw can respond normally, enter script generation specific testing.

Terminal input to enter TUI:

```
openclaw tui

openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)

|
```

It is recommended to start with the simplest script requirements, for example:

- Help me write a script to make RGB red light blink 5 times, save to desktop and run
- Generate a script to read temperature and humidity, don't run first
- Write a script to make servo 1 swing back and forth between 0, 90, 180 degrees
- Generate a script for OLED to display "Hello OpenClaw"

## 5. Three Solutions for Communicating with OpenClaw

| Solution | Suitable Scenarios                                   | Advantages                                          | Recommended Positioning                                    |
|----------|------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------|
| whatsapp | Remote requirement sending, classroom demonstrations | Can directly describe "what script to write for me" | Used to demonstrate natural language generation capability |
| TUI      | Local debugging, verify generated results            | Easiest to see script generation and run returns    | Preferred debugging solution                               |

| Solution   | Suitable Scenarios                        | Advantages                                         | Recommended Positioning            |
|------------|-------------------------------------------|----------------------------------------------------|------------------------------------|
| Voice code | Voice interaction, complete AI experience | Can directly verbally describe script requirements | As complete demonstration solution |

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

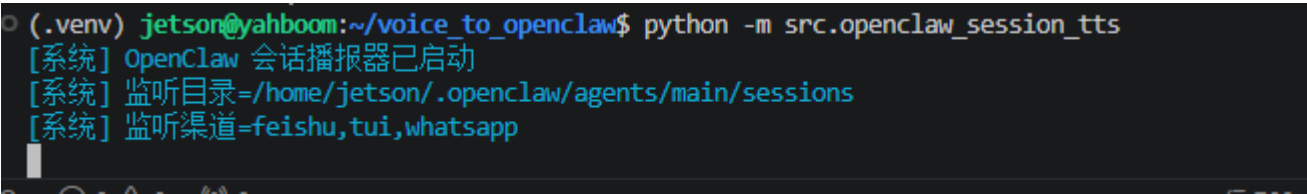
- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



## 5.1 whatsapp Solution

For whatsapp connection steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. This section only retains dialogue examples suitable for script generation tutorials.

You can directly dialogue, for example:

- Generate a script that make servo swing left and right for 10 seconds and run it

20:07



< 2



+86 153 3885 7526 (你)

给自己发消息

HTTP 400: Access denied, please make sure your account is in good standing. For details, see: <https://help.aliyun.com/zh/model-studio/error-code#overdue-payment>

19:59 ✓✓

Generate a script that makes servo swing left and right for 10 seconds and run it

20:07 ✓✓

Done! Generated `servo_swing.py` on the desktop and executed it successfully.

The servo swung left ( $0^\circ$ ) and right ( $180^\circ$ ) repeatedly for 10 seconds, then returned to the center position ( $90^\circ$ ).

20:07 ✓✓



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space

return

## 5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter the following command in the terminal to continue the dialogue:

```
openclaw tui
```

Then you can directly enter script requirements, for example:

- Help me write a script to control RGB to become blue breathing light and run it

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
  
session agent:main:main  
connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

## 5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

- Write a script that displays current temperature and humidity to the screen and run it

```
[系统] 串口设备已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Write a script that displays current temperature and humidity on the screen and run it.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! I created and ran the script `temp_humidity_display.py` on your desktop. 📄

The script:
1. Read the current temperature:
    **30.3°C**
2. Read the current humidity: **50.0%**
3. Displayed both
    values on the OLED screen for 15 seconds
4. Cleared the screen
    automatically

The display is now cleared. The script is saved on
your desktop if you want to run it again later!
[信息] 回复已裁剪为播报文本: 372 -> 76 字
```

## 6. Comprehensive Cases

Note: The following are all dialogue demonstrations using `openclaw_tui` method. You can also choose whatsapp or voice method to complete.

### 6.1 Case One: RGB Blinking Script

#### Experimental Objective

Let OpenClaw generate a basic light script to verify the "script creation + peripheral execution" chain.

#### Effect Description

OpenClaw will organize your natural language requirements into a Python script. After the script is saved to the desktop, it can be run directly or kept for later modification.

#### Example Instructions

- Help me write a script to make RGB red light blink 5 times
- Generate a blue breathing light script, save to desktop and run
- Write a script for RGB to make a rainbow light effect

## 6.2 Case Two: Servo Action Script

### Experimental Objective

Generate a servo action script to solidify single dialogue actions into repeatable programs.

### Effect Description

This type of script is suitable for actions like "nodding", "shaking head", "swinging back and forth", "returning to center", and is also suitable for later expansion into classroom demonstrations or combined projects.

### Example Instructions

- Write a script for servo 1 to reciprocate between 0, 90, 180 degrees
- Generate a servo nodding script
- Help me write a script for four servos to all return to 90 degrees

## 6.3 Case Three: Temperature and Humidity Reading Script

### Experimental Objective

Generate a script that reads current temperature and humidity and outputs results.

### Effect Description

This type of script is suitable for sensor self-check, data sampling, or subsequent log recording. It is also the most commonly used basic script type when making scheduled tasks later.

### Example Instructions

- Generate a script to read temperature and humidity, save to desktop
- Help me write a script to print current temperature and humidity
- Write a script to read DHT data once and run it

## 6.4 Case Four: OLED Display Script

### Experimental Objective

Generate a script that displays text or indicator content to OLED.

### Effect Description

This case is suitable for making welcome pages, status pages, simple dashboards, and is also the foundation for later linking with capabilities like temperature/humidity and cameras.

### Example Instructions

- Help me write a script for OLED to display "Hello OpenClaw"
- Generate a script to display temperature and humidity to OLED
- Write a script to display a simple status information page

## 6.5 Case Five: Combined Script

### Experimental Objective

Let OpenClaw generate a comprehensive script containing multiple peripheral actions at once.

### Effect Description

This type of script can chain together "reading sensors", "displaying to OLED", "controlling RGB", "driving servos", suitable for transitioning from single peripheral control to small project scripts.

### Example Instructions

- Write a comprehensive script: read temperature and humidity then display to OLED
- Generate a script: first light green, then turn servo to 90 degrees
- Help me write a script: if temperature is too high, light red and prompt on OLED